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Speech-Based Alzheimer's Disease Detection Using Bayesian Optimized Fusion of SVM and Random Forest

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Abstract

Classification of Alzheimer's disease through speech utilizing the feature set is quite efficient. This paper proposes a method related to developing a classifier using machine learning algorithms and some parameters including MFCC, ZCR, spectral centroid, and chroma.

The analysis of the feature set leads to creating the reference classifiers using SVM with RBF kernel and random forest (RF). To achieve more accurate results, fusion techniques can be used. In particular, the output generated by the classifiers, which include SVM and RF, will act as inputs for the metaclassifier employing Bayesian optimization.

From the experimental results, it becomes clear that applying the fusion approach allows attaining better results than other approaches that employ other classifiers. Accuracy, precision, recall, F1 score, and confusion matrix have been considered as validation metrics.

It should be pointed out that the newly created optimized technique makes it possible to increase the efficiency of classification.

Keywords: Alzheimer's Disease; Speech Analysis; Machine Learning; Support Vector Machine; Random Forest; Bayesian Optimization; Feature Extraction; MFCC; Ensemble Learning; Fusion Models

Introduction

Firstly, we consider the neurodegenerative disease affecting memory, which adversely impacts the patient's daily routine activities, namely, Alzheimer's disease. The primary disadvantage of the current procedure for diagnosing Alzheimer's disease is the cost of such tests and the time it takes to conduct them.

Many research works have been conducted in order to assess the possibility of using speech as the method for diagnosing Alzheimer's disease by analyzing articulation, fluency, and prosody. Acoustic feature extraction could be regarded as a vital step in designing such applications. The most frequently utilized acoustic features are MFCC, ZCR, Spectral Centroid, and Chroma. Different algorithms for classifying problems concerning the abovementioned topic have been reviewed, such as SVM and RF. Nevertheless, despite some considerable success, the problem of robustness has become evident.

To solve the given problem, ensemble learning and fusion can be used. In fact, there are numerous algorithms utilizing the most basic one, which consists of classification of the fused output and averaging the input data. The contribution of the study to the field is associated with the design of an innovative system for the diagnosis of the disease using the recognition of the speech via parallel classifiers and Bayesian Optimization. The innovation introduced by the research relates to its use of an innovative system of combination of classifiers' outputs based on a meta-classifier, which turned out to be effective. The effectiveness of the proposed system was estimated according to the following parameters: accuracy, precision, recall, F1-score, and confusion matrix.

Literature Review

The application of speech processing methods has facilitated the development of an ingenious technique where the presence of Alzheimer's disease can be detected through voice analysis. As pointed out by De Filippis and Al Foysal [1], it is possible to detect the presence of Alzheimer's disease efficiently and easily through language-based features that have been extracted from brief speech samples. Although the method suggested is very efficient, it does not take advantage of any of the merits offered by acoustic features since it only considers the linguistic attributes.

According to Kiyoshige et al. [2], the speech signal provides valuable insight into diagnosing the Alzheimer's disease at an initial stage.

In contrast, Sharafeldeen et al. [4] highlighted the importance of machine learning algorithms such as the SVM and ensemble methods in detecting Alzheimer's disease using speech samples, along with exploring the problems involved with generalization and variance in the data set.

Finally, Oiza-Zapata et al. [5] pointed out the significance of interpretability in using machine learning algorithms for medicine, where interpretability is vital when working in the area of diagnostics.

In addition, several researchers such as Wang et al. [10] used various multimodal approaches, where speech signals were combined with other data sets to increase detection rates, but this also leads to increased system complexity.

However, despite all these advancements, the dependency of modern-day techniques on either single classifiers or complex approaches remains important. Only a very few number of techniques are found which are based on the combination of appropriate classifiers through fusion techniques.

Limitations

Although the importance of integrating the SVM and random forest in enhancing the performance of the algorithm that diagnoses Alzheimer's disease cannot be underestimated, there are some difficulties resulting from the application of the acoustic data. The reason for this is twofold; firstly, acoustic data is susceptible to any change in sound recording or sound, and secondly, due to the fact that data analysis was conducted on a limited amount of data.

Research Gap

However, despite the remarkable progress that has been made towards using voice recognition techniques for identifying Alzheimer's disease, the systems used today utilize either the single classifier system or the complex deep learning algorithms. The single classifier system is unable to account for numerous decision-making processes and thus lacks generalization ability, whereas the complex algorithms require vast amounts of data and computation. In addition, the fusion systems used today only utilize the averaging or majority voting schemes and do not take advantage of the individual classifier strengths.

Proposed Approach

The methodology for detecting Alzheimer's disease has been tested by applying it to extract the speech signal in the diagnosis of Alzheimer's disease according to the pipeline that includes preprocessing, feature extraction, classification, and optimization of fusion.

Proposed Solution

The proposed solution involves the application of the Bayesian optimization algorithm in the development of a fusion algorithm to enhance the model's efficiency. Specifically, the Bayesian optimization algorithm will be utilized to fuse the outputs generated by support vector machine (SVM) and random forest algorithms. Here, the outputs of the support vector machine algorithm will provide decision values while the random forest will provide class probabilities. The outputs generated by these two classifiers will then be applied to train the logistic regression model as a meta classifier.

In this respect, the proposed solution involves the utilization of the Bayesian optimization algorithm to balance the two classifiers, namely, the support vector machine algorithm and the random forest algorithm.

Methodology

Our suggested system architecture is intended for diagnosing patients with Alzheimer's disease via analysis of speech signals and machine learning approach. Four phases that constitute the proposed scheme include:

i. Speech Preprocessing,
ii. Feature extraction,
iii. Classification using multiple classifiers, and
iv. Decision level fusion based on Bayesian optimization algorithm.

System Architecture

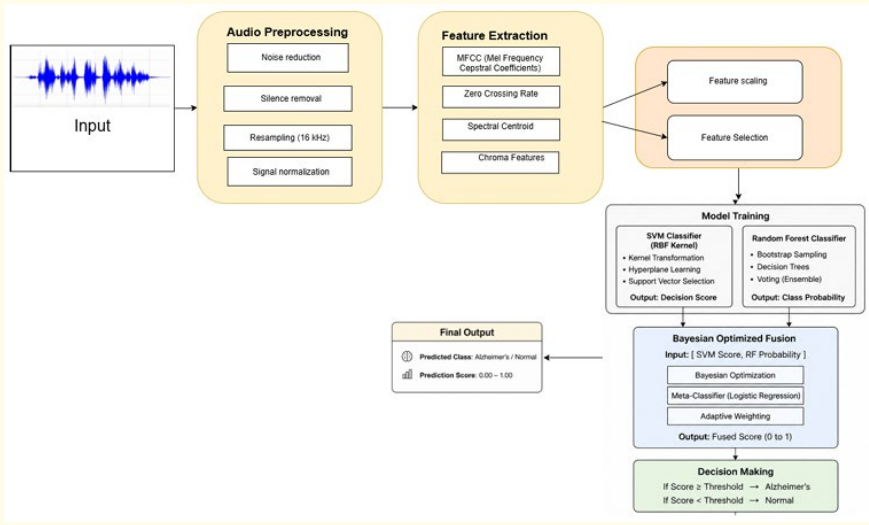


Figure 1: Speech-Based Alzheimer's Detection System Architecture.

The designing of Alzheimer's disease detection architecture is discussed in this section, where a pipeline architecture is adopted in which audio signals are pre-processed using noise filtering, noise reduction, and noise normalization techniques. The feature extraction phase involves majorly Mel Frequency Cepstral Coefficients, Zero Crossing Rate, Spectral Centroid, and Chroma feature extraction techniques. Machine learning classifiers like SVM and RF classifiers are used to implement the fusion layer of classifier by adopting Bayesian optimization technique in order to develop the metaclassifier algorithm. Finally, decision making algorithm is adopted for the classification of Alzheimer's disease.

Speech Preprocessing

The input to the speech signal is denoted as $x(t)$, with t being time. Afterward, there is filtering of the signal for noise and silences within it. The next process involves resampling of the signal at a fixed sampling frequency of 16 kHz.

Feature Extraction

The feature extraction stage entails finding features related to cognitive impairment in the processed speech signal. The feature vector F is denoted as:

$$F = [MFCC_1, MFCC_2, \ldots, MFCC_n, ZCR, SC, CF]$$

Where in, MFCC stands for Mel Frequency Cepstral Coefficients, ZCR is the Zero Crossing Rate, SC refers to Spectral Centroid, while CF is Chroma Feature.

Feature Scaling

In order to have an even distribution of features, the method of feature vector extraction follows the equation given below:

$$X' = \frac{X - \mu}{\sigma}$$

Where μ and σ are the mean and standard deviation of the training data set.

Classification Models

Below, two types of machine learning models are used together in order to make decision boundaries for varying patterns within the feature space.

Support Vector Machine (SVM): Machine learning algorithms are considered for use together so as to determine the decisions on the basis of the patterns generated by the features.

The algorithm for solving the problem of non-linear separability is the SVM algorithm with the help of the RBF kernel for recognizing the decision boundary. The decision function can be written as given below:

$$f(x) = \sum_{i=1}^{n} \alpha_i y_i K(x_i, x) + b$$

Random Forest (RF): The random forest method can be regarded as one of the methods involving building several decision trees with the help of bootstrapping techniques. The result obtained is determined by averaging the result obtained in every single decision tree. It can be mathematically expressed as follows:

$$\hat{y} = \frac{1}{N} \sum_{k=1}^{N} T_k(x)$$

Here, $T_k(x)$ stands for the output obtained in the k-th decision tree.

Bayesian Optimized Fusion

To improve the performance of the classification problem, the decision level fusion method is used to combine the output of the SVM and Random Forest techniques. In contrast to simply averaging the outcomes, the presented technique employs a meta-classifier that decides on the optimal combination of the model outputs.

Let P_{svm} and P_{rf} denote the output obtained from SVM and Random Forest. The fused outcome can be represented as

$$P_{fusion} = w_1 P_{svm} + w_2 P_{rf}$$

The parameters w_1 and w_2 are obtained by using Bayesian optimization. The meta-classifier adopted here is the logistic regression model.

Model Evaluation

The proposed model's performance is analyzed using traditional performance measures including accuracy, precision, recall, and F1-score. In addition, the use of confusion matrix techniques is considered in assessing the model's performance and error analysis.

Results and Analysis

Model Performance Evaluation

SVM Performance: The evaluation of the effectiveness of the SVM algorithm was done by using the confusion matrix approach. As seen in Fig. 2, the algorithm successfully identified 244 instances as normal and 124 instances as Alzheimer. Nevertheless, there were 34 instances that could not be classified as normal, and 28 instances were incorrectly classified as Alzheimer.

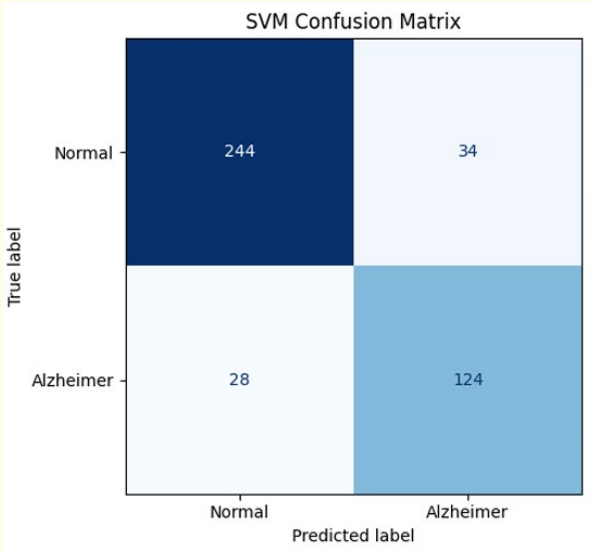


Figure 2: Confusion Matrix of SVM Model.

The SVM classifier accuracy score is 0.856, which means that the performance of the classifier in terms of classification is quite average. From the large number of errors committed by the classifier, we can conclude that the SVM classifier lacks the ability to classify the classes when required.

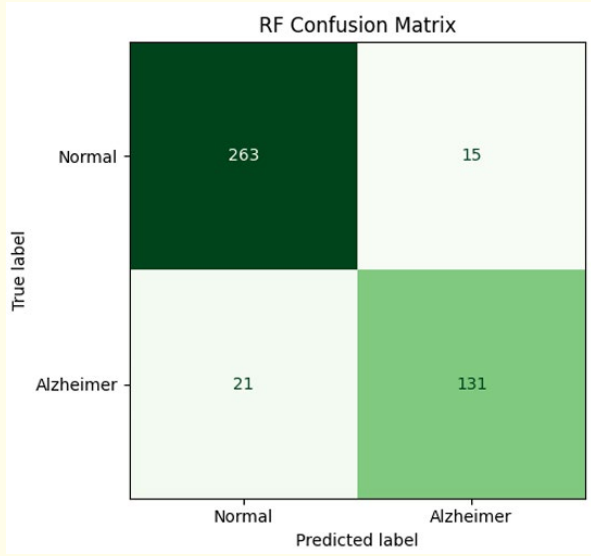


Figure 3: Confusion Matrix of Random Forest Model.

Random Forest Performance: Assessment of the Random Forest (RF) classifier was conducted using the confusion matrix approach. From the Fig. 3, it can be observed that the RF classifier classifies 263 samples in the normal category and 131 samples in the Alzheimer's category. However, it wrongly classifies 15 samples of the normal category into the Alzheimer's category and 21 samples of the Alzheimer's category into the normal category.

From the above analysis, it can be concluded that the accuracy level of the RF classifier is 0.916. The RF classifier outperforms the SVM classifier in terms of accuracy. It means that there are few errors made in classification since the RF classifier uses an ensemble approach to uncover relationships among the features within the feature space.

Fusion Model Performance: The analysis of the fused model has been done with the help of the output of SVM and random forest by applying the confusion matrix method. The performance analysis of the model from the results generated using this technique indicates that the fused model has classified 264 healthy and 135 cases of Alzheimer's disease patients and wrongly classified 14 healthy and 17 Alzheimer's disease patients as illustrated in Fig. 4.

The fused model demonstrates a high level of accuracy of 0.895, indicating that the fusion helps to minimize the errors produced in the prediction process by different classifiers. Therefore, errors are minimized due to the fusion. Unfortunately, it does not perform better than the Random Forest classifier concerning the accuracy value.

Therefore, it can be concluded that fusion alone does not improve the performance; hence, more advanced methods like Bayesian Optimization have to be adopted.

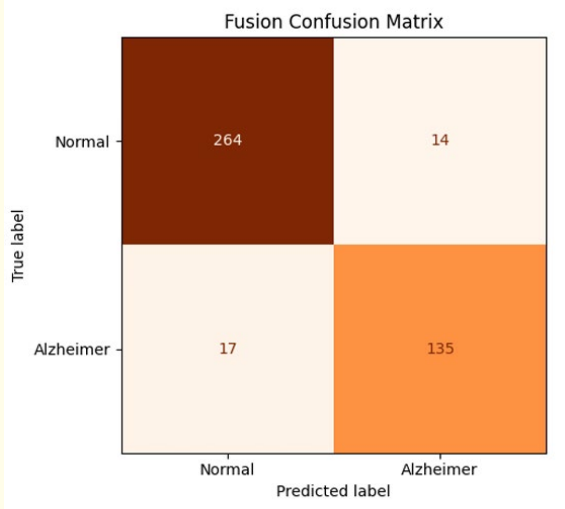


Figure 4: Confusion Matrix of Fusion Model.

Model Comparison

All the models were compared and this can be seen from Fig. 6. The SVM model had an accuracy of 0.856, which indicates that although it provides reasonable accuracy, there is room for improvement since the errors involved are rather high. To increase the accuracy to 0.916, the RF model was used.

The accuracy score for the baseline fusion model was expected to be 0.895, meaning that it performed really well but did not outperform the RF model since it managed to balance the error rates. However, the optimization process conducted on the fusion model showed excellent results by providing the highest accuracy score of 0.928.

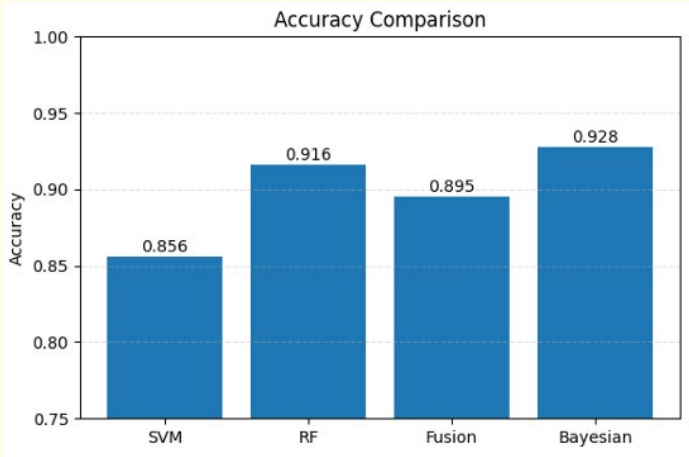


Figure 5: Accuracy Comparison.

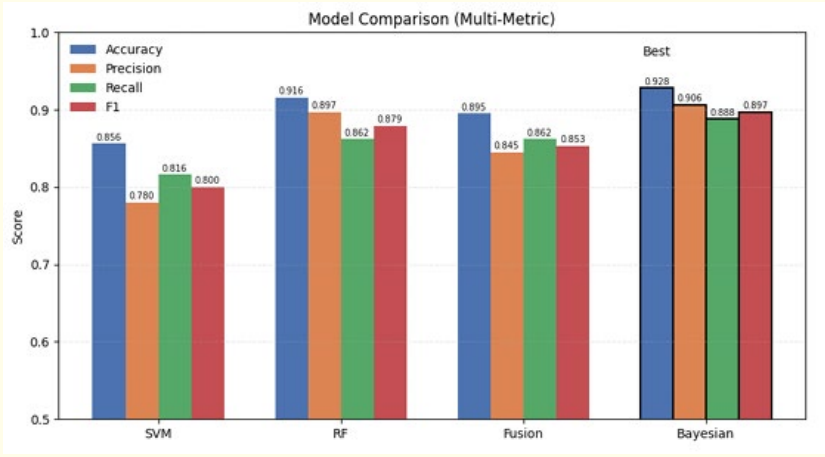


Figure 6: Performance comparison of SVM, RF, Fusion, and Bayesian Fusion models.

Based on Table 1, it is clear that Bayesian optimization of the fusion model gave the best scores for the precision score of 0.906, recall score of 0.888, and F1 measure of 0.897.

Analysis of Feature Representation

In figure 7, the MFCC representation of the speech signal is illustrated. Heat Map shows the changes in the cepstral coefficients with regard to time. Lower order cepstral coefficients include the complete spectral representation of the signal, while the higher order cepstral coefficients include the variations in the speech signals.

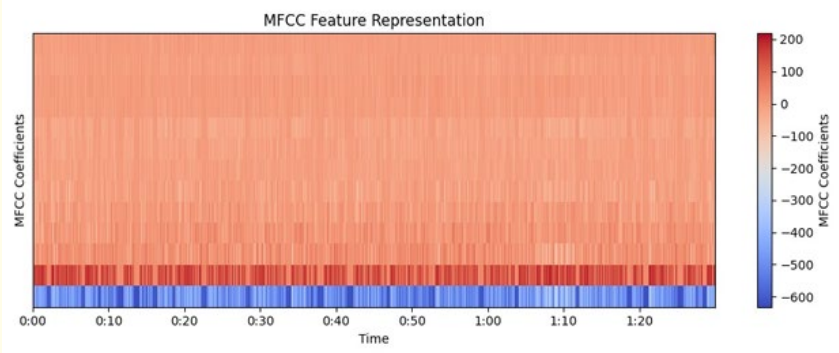


Figure 7: Feature representation.

Discussion

From the above experimental findings, efficiency differs from one technique to another used in detecting Alzheimer's disease. SVM serves as the baseline model that fails to overcome the challenge of misclassification. However, RF becomes highly efficient because of adopting an ensemble learning strategy. Despite the fusion model becoming more efficient in resolving the issues of false positive/

false negatives than SVM, it becomes less efficient than RF. As such, the adaptive fusion model becomes much more efficient than other models since it adopts the Bayesian decision-making strategy.

Model	Accuracy	Precision	Recall	F1-Score
SVM	0.856	0.780	0.816	0.800
Random Forest	0.916	0.897	0.862	0.879
Fusion	0.895	0.845	0.862	0.853
Bayesian Fusion	**0.928**	**0.906**	**0.888**	**0.897**

Table 1: Performance Comparison of Models.

Conclusion

The proposed research study introduces a novel methodology for the effective detection of speech characteristics of patients with Alzheimer's disease through the application of acoustic characteristics and machine learning models. Particularly, the combination of SVM and RF classifiers was considered, and the decision-level fusion methodology was applied as the means of integrating them into one model.

As proven in the conducted experiments, the RF classification method outperforms the SVM technique in terms of classification performance; however, the implementation of decision-level fusion did not lead to sufficient results comparing to the RF classifier. To cope with the issue, optimization of the machine learning algorithm combination was carried out on the basis of Bayesian optimization, which led to achieving the maximum accuracy rate of 0.928.

Thus, from the results presented, one can see that optimization of machine learning techniques and combining them resulted in improving their efficiency. The current research study, thus, contributes to the advancement of speech recognition methodologies in Alzheimer's disease diagnosis.

In conclusion, it should be noted that combination and optimization of diverse machine learning approaches increase their efficiency.

Future Scope

However, there are several ways to further improve the proposed speech analysis-based algorithm for Alzheimer's disease recognition. First, by increasing the size of the dataset and incorporating greater diversity in the training data, one can make sure that the generalizability and the bias of the algorithm prediction will be higher. Another way to enhance the performance of the algorithm might include introducing additional information concerning the temporal features of the voice and incorporating such features as the duration between consecutive words, speed, and articulation into the feature vector.

Moreover, incorporating linguistic information, or any kind of clinical data into the analysis could also help increase the accuracy of prediction. Finally, incorporating modern speech recognition algorithms, such as neural networks for voice analysis trained on large amounts of data might further increase the accuracy of the predictions made by the algorithm. In what concerns the technical aspects of the implementation, one also needs to consider increasing efficiency in real-time computations and decreasing the computational complexity of the method.

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